

Connect MongoDB

```
C:\Users\ELCOT>mongosh
```

```
Current Mongosh Log ID: 6ac120ee8665a54258290cda
```

```
Connecting to:
```

```
mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
```

```
Using MongoDB:      9.0.2
```

```
Using Mongosh:      2.12.0
```

```
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
```

```
-----
```

```
The server generated these startup warnings when booting
```

```
2026-10-02T09:31:08.615+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
```

```
-----
```

```
test> use studentDB
```

```
switched to db studentDB
```

```
studentDB> db.createCollection('students')
```

```
{ ok: 1 }
```

```
studentDB> show collections
```

```
students
```

```
C:\Users\ELCOT>mongosh
Current Mongosh Log ID: 6abfdbef733ff3aa3af94050
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB:      9.0.2
Using Mongosh:      2.12.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-10-02T09:31:08.615+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin    40.00 KiB
config  96.00 KiB
local   72.00 KiB
test> use studentDB
switched to db studentDB
studentDB> db.createCollection('students')
{ ok: 1 }
studentDB> show collections
students
studentDB> db.students.insertOne({name:'Selva', age:19, course:'MERN Stack', status:'Almost done'})
{
  acknowledged: true,
  insertedId: ObjectId('6abfdc84733ff3aa3af94051')
}
```

Insert one data

```
studentDB> db.students.insertOne({name:'Selva', age:19, course:'MERN Stack', status:'Almost done'})
{
  acknowledged: true,
  insertedId: ObjectId('6ac121358665a54258290cdb')
}
```

```
studentDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  }
]
```

```
studentDB> db.students.insertOne({name:'Selva', age:19, course:'MERN Stack', status:'Almost done'})
{
  acknowledged: true,
  insertedId: ObjectId('6abfdc84733ff3aa3af94051')
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  }
]
```

Insert many data

```
studentDB> db.students.insertMany([ {name:'Suriya', age:21, course:'Java', status:'Completed'},
{name:'Sathish', age:20, course:'MERN Stack', status:'Ongoing'}, {name:'Gokul', age:23, course:'Python',
status:'Completed'}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac122858665a54258290cdc'),
    '1': ObjectId('6ac122858665a54258290cdd'),
    '2': ObjectId('6ac122858665a54258290cde')
  }
}
```

```
studentDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',

```

```

age: 21,
course: 'Java',
status: 'Completed'
},
{
  _id: ObjectId('6ac122858665a54258290cdd'),
  name: 'Sathish',
  age: 20,
  course: 'MERN Stack',
  status: 'Ongoing'
},
{
  _id: ObjectId('6ac122858665a54258290cde'),
  name: 'Gokul',
  age: 23,
  course: 'Python',
  status: 'Completed'
}
]

```

```

studentDB> db.students.insertMany([{'name':'Suriya', age:21, course:'Java', status:'Completed'}, {'name':'Sathish', age:20, course:'MERN Stack', status:'Ongoing'}, {'name':'Gokul', age:23, course:'Python', status:'Completed'}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abfdd97733ff3aa3af94052'),
    '1': ObjectId('6abfdd97733ff3aa3af94053'),
    '2': ObjectId('6abfdd97733ff3aa3af94054')
  }
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Ongoing'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94054'),
    name: 'Gokul',
    age: 23,
    course: 'Python',
    status: 'Completed'
  }
]

```

Find particular data

```

studentDB> db.students.find({course:'MERN Stack'})
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',

```

```

    status: 'Almost done'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Ongoing'
  }
]

```

```

studentDB> db.students.find({course: 'MERN Stack'})
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Ongoing'
  }
]

```

Update one data

```

studentDB> db.students.updateOne({name: 'Sathish'}, {$set: {status: 'Almost done'}})

```

```

{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

```

```

studentDB> db.students.find()

```

```

[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,

```

```

course: 'Java',
status: 'Completed'
},
{
  _id: ObjectId('6ac122858665a54258290cdd'),
  name: 'Sathish',
  age: 20,
  course: 'MERN Stack',
  status: 'Almost done'
},
{
  _id: ObjectId('6ac122858665a54258290cde'),
  name: 'Gokul',
  age: 23,
  course: 'Python',
  status: 'Completed'
}
]

```

```

studentDB> db.students.updateOne({name: 'Sathish'}, {$set: {status: 'Almost done'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Almost done'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94054'),
    name: 'Gokul',
    age: 23,
    course: 'Python',
    status: 'Completed'
  }
]

```

Update many data

```
studentDB> db.students.updateMany({course:'MERN Stack'}, {$set:{status:'Completed'}})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
studentDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cde'),
    name: 'Gokul',
    age: 23,
    course: 'Python',
    status: 'Completed'
  }
]
```

```

studentDB> db.students.updateMany({course:'MERN Stack'}, {$set:{status:'Completed'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94054'),
    name: 'Gokul',
    age: 23,
    course: 'Python',
    status: 'Completed'
  }
]

```

Delete one data

```

studentDB> db.students.deleteOne({name:'Gokul'})
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.find()
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),

```

```
name: 'Sathish',
age: 20,
course: 'MERN Stack',
status: 'Completed'
}
]
```

```
studentDB> db.students.deleteOne({name: 'Gokul'})
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

\$gt, \$lt and \$in

```
studentDB> db.students.find({age: {$gt:20}})
```

```
[
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  }
]
```

```
studentDB> db.students.find({age: {$lt:20}})
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

```
studentDB> db.students.find({course:{$in:['MERN Stack', 'Java']}})
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

```
studentDB> db.students.find({age:{$gt:20}})
[
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  }
]
studentDB> db.students.find({age:{$lt:20}})
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
studentDB> db.students.find({course:{$in:['MERN Stack', 'Java']}})
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

\$and , \$or

```
studentDB> db.students.find({$and:[{age:{$gt:20}}, {status:'Completed'}]})
```

```
[
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  }
]
```

```
studentDB> db.students.find({$or:[{course:'MERN Stack'}, {course:'Python'}]})
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

```
studentDB> db.students.find({$and: [{ age: { $gt: 20 } }, { status: "Completed" } ]})
[
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  }
]
studentDB> db.students.find({$and:[{age:{$gt:20}}, {course:'Python'}, {status:'Completed'}]})
studentDB> db.students.find({$or:[{course:'MERN Stack'}, {course:'Python'}]})
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
```

\$exists

```
studentDB> db.students.updateOne({name:'Selva'}, {$set:{email:'selva@gmail.com'}})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
studentDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Complated',
    email: 'selva@gmail.com'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdc'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6ac122858665a54258290cdd'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Complated'
  }
]
```

```
studentDB> db.students.find({email:{$exists:true}})
```

```
[
  {
    _id: ObjectId('6ac121358665a54258290cdb'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Complated',
    email: 'selva@gmail.com'
  }
]
```

```

studentDB> db.students.updateOne({name:'Selva'}, {$set:{email:'selva@gmail.com'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed',
    email: 'selva@gmail.com'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94052'),
    name: 'Suriya',
    age: 21,
    course: 'Java',
    status: 'Completed'
  },
  {
    _id: ObjectId('6abfdd97733ff3aa3af94053'),
    name: 'Sathish',
    age: 20,
    course: 'MERN Stack',
    status: 'Completed'
  }
]
studentDB> db.students.find({email:{$exists:true}})
[
  {
    _id: ObjectId('6abfdc84733ff3aa3af94051'),
    name: 'Selva',
    age: 19,
    course: 'MERN Stack',
    status: 'Completed',
    email: 'selva@gmail.com'
  }
]

```

Delete many data

```
studentDB> db.students.deleteMany({})
```

```
{ acknowledged: true, deletedCount: 3 }
```

```
studentDB> db.students.find()
```

```
studentDB>
```

```

studentDB> db.students.deleteMany({})
{ acknowledged: true, deletedCount: 3 }
studentDB> db.students.find()

studentDB>

```